

P100 Unit Diagnostic Report

Disk / Storage Failure - Unit Does Not Boot

Node (planename / tail ID)	N158AV
Unit Type	KONUK
Serial Number	KONUK10295945
SKU Number	P100-3211022
Diagnostic Date	August 20, 2026
Diagnostic Method	SystemRescueCD (live USB boot)
Diagnosis	Hardware failure - NVMe storage disk not detected
Symptom	Unit does not boot into OS

Diagnostic Summary

This unit fails to boot into its operating system. Following standard restore procedure, the unit was booted from a SystemRescueCD live USB to inspect the internal storage disk. From the live environment, the internal NVMe disk is not visible at any level: it does not appear in the block device list, is not detected on the PCI bus, produces no kernel messages on enumeration attempts, and does not appear in the NVMe management tool's device list. Only the boot USB itself is visible to the system. This is consistent with a hardware-level storage failure - either a dead NVMe controller or a physical seating/connection issue that could not be ruled out due to lack of physical access to the unit during this diagnostic session.

Troubleshooting Steps Performed (in order)

1. Booted the unit from SystemRescueCD live USB, following the standard P100 restore procedure, since the unit does not boot into its normal OS.
2. Checked visible block devices (`lsblk -l`): only the boot USB itself (`sda`, with its boot partition and rescue partition) is listed. No internal disk (expected to appear as `nvme0n1`) is present.
3. Checked the PCI bus for an NVMe controller (`lspci | grep -i nvme`): no NVMe device found on the PCI bus at all.
4. Checked kernel boot/device messages for any NVMe-related activity (`dmesg | grep -i nvme`): no output - the kernel never attempted to enumerate an NVMe device.
5. Queried the NVMe management utility directly (`nvme list`): returned an empty device table, confirming no NVMe device is registered with the system.
6. Physical inspection of the M.2/NVMe module and its seating in the slot could NOT be performed during this session (no physical access to the unit at the time of diagnosis). This is the one remaining step that could distinguish between a dead controller and a loose/disconnected module, and is recommended before or during hardware replacement.

Conclusion: the internal NVMe disk is not detected at any level of the system (block device, PCI bus, kernel messages, or NVMe management tool), which is consistent with a hardware-level storage failure. Physical inspection of the module's seating is recommended as a first step upon physical access to the unit; if the module is properly seated and the disk is still not detected, disk replacement is recommended.

Appendix: Technical Evidence (logs and command output)

Unit Identification (dmidecode)

Planename / Tail ID: N158AV
Serial Number: KONUK10295945
UUID: 03000200-0400-0500-0006-000700080009
Wake-up Type: Power Switch
SKU Number: P100-3211022

Note: the serial number, UUID, and SKU were obtained via dmidecode (chassis/board-level data), which remains available even though the OS disk itself could not be read. The planename/tail ID (N158AV) was provided separately.

Block Devices (lsblk -l)

```
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
loop0 7:0 0 673.7M 1 loop /run/archiso/sfs/airootfs
sda 8:0 0 953.9G 0 disk
sda1 8:1 0 1.8G 0 part /run/archiso/bootmnt
sda2 8:2 0 892.4G 0 part
```

Note: "sda" here is the SystemRescueCD boot USB itself (sda1 is mounted as the rescue boot partition). No internal NVMe disk is listed.

PCI Bus - NVMe Controller Check (lspci | grep -i nvme)

(no output - no NVMe controller found on the PCI bus)

Kernel Messages - NVMe Enumeration (dmesg | grep -i nvme)

(no output - the kernel produced no NVMe-related messages during boot)

NVMe Device List (nvme list)

Node	Generic	SN	Model	Namespace	Usage	Format	FW	Rev

(empty table - no NVMe devices registered with the system)